The idea isn't as far-fetched as it sounds. There's a principle of statistical mechanics that says that any arrangement a system of particles finds itself in can happen again, if you wait long enough. Let's say you have a box filled with a gas of randomly moving molecules, and you take a snapshot of them at one moment and mark down what positions they're in. If you watch the box for a very long time, eventually you'll find the molecules in those positions again. The less likely the configuration, the longer it'll take, so a very rare event like all the particles huddling up in the lower right-hand corner of the box will take a lot longer to appear again, but in principle, it's just a matter of time. This is called *Poincaré recurrence*. If you have an infinite amount of time to work with, any state the system can be in is a state it WILL be in again, an infinite number of times, with a recurrence time determined by how rare or special that configuration is. In one rather arresting example, physicists Anthony Aguirre, Sean Carroll, and Matthew Johnson once calculated that if you were willing to wait something like a trillion trillion times the age of the universe, you could watch an entire piano spontaneously assemble itself in a seemingly empty box.

A post–Heat Death universe is, essentially, a very large, very slightly warmed box, with statistical mechanics stepping in to provide the random fluctuations. If the Big Bang is a state the universe has been in once, and the post–Heat Death universe is eternal (so eternal that, having lost the arrow of time, past and future are meaningless), there's no reason a Big Bang can't fluctuate out of the vacuum to start the universe anew.

Hold on, though. It gets even weirder. And more personal.

If *every* state the universe has ever been in could be revisited through random fluctuations, that means *this moment right now* could happen again, exactly the same in every detail. Not only could it happen again, it could happen again *infinitely many times*.

This possibility is of particular interest to cosmologist Andreas Albrecht, who has written about what he calls the *de Sitter Equilibrium* state. The basic idea of this equilibrium version of de Sitter space is that the origin of our universe and everything that happens in it can be thought of as the result of random fluctuations out of an eternally expanding universe containing only a cosmological constant. From time to time, a universe fluctuates out of the heat bath into a very low entropy starting state, and then evolves forward (with increasing entropy) until it gets to its own Heat Death, decaying back into the background de Sitter universe. And from